# Assignment #1: Fingerprint Recognition

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## I. Introduction

We perform and evaluate fingerprint verification with the NIST Biometric Image Software (NBIS). Minutiae are extracted with `mindtct`, compared with `bozorth3` and image quality is estimated with `nfiq`. We also test whether pre-classifying fingerprints by pattern type with `pcasys` improves verification.

## II. Methodology

We process each fingerprint with NBIS tools. `mindtct` extracts the minutiae, `bozorth3` compares the minutiae of two fingerprints and returns a similarity score, and a threshold $T$ turns the score into a decision: a pair is genuine (same finger) if its score is $\geq T$ and impostor otherwise.

Independently of the matching, `nfiq` rates the quality of each image from 1 (best) to 5 (worst). The rating does not influence the decision, but it helps us explain low scores.

We select $T$ from the scores of all pairs. For every candidate threshold, we compute the false acceptance rate (FAR), the share of impostor pairs accepted as genuine and the false rejection rate (FRR), the share of genuine pairs rejected. The equal error rate (EER) is the point where FAR equals FRR. Then we also compute the accuracy, the share of correctly classified pairs. We choose the threshold with the highest accuracy, because accuracy is our evaluation measure and impostor pairs outnumber genuine pairs, which moves the best threshold above the EER threshold.

A second experiment sorts the fingerprints by pattern type before matching. `pcasys` predicts one of six types. We match only pairs with the same predicted type and label all other pairs as impostor without matching.

## III. Experiments with Results

We used the FVC2000 DB1 B dataset which gives 80 plain optical images of 300×300 px at 500 dpi. Images were converted from TIFF to PNG and minutiae computed with `mindtct`. Every pair was compared once with `bozorth3` (no self or repeated comparisons), giving 3160 pairs: 280 genuine and 2880 impostor. To choose $T$, we swept it from 0 to 250 and computed FAR, FRR and accuracy. We report the EER threshold, but select the threshold with the highest accuracy, because accuracy is our evaluation measure and the classes are imbalanced. In the second experiment, images were upscaled to 600×600, converted to WSQ (`cwsq`, 0.75 bit rate) and typed by `pcasys` into A, L, R, S, T or W. Pairs with different predicted types are declared different subjects without matching. Pairs of the same type are matched with `bozorth3` as before.

Impostor scores lie mostly below 25, while genuine scores spread up to 245 and overlap with impostors at low values (Fig. 1).

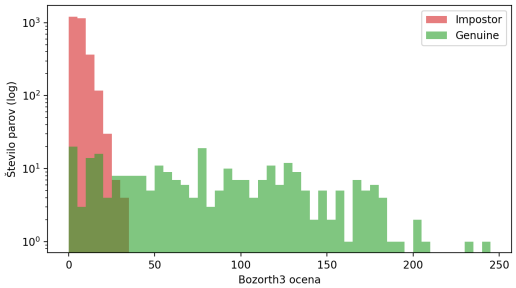



Fig. 1. Bozorth3 scores of genuine and impostor pairs (log y-axis).

The similarity matrix shows bright diagonal blocks, one per finger (Fig. 2).

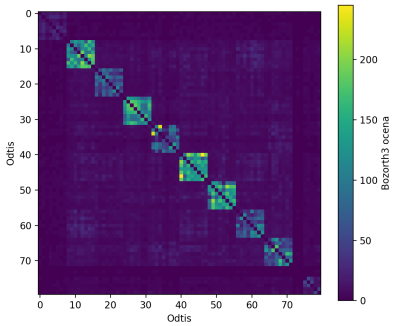



Fig. 2. Similarity matrix of all 80 images; bright diagonal blocks are the 8 impressions of one finger.

Quality is mostly good (Fig. 3): NFIQ 1/2/3/4/5 for 24/30/23/0/3 images. This is also reflected in the similarity matrix, where image quality helps explain the weaker blocks. Finger 102 has the best mean NFIQ score (1.0),

indicating great quality across all its fingerprints, and its block is the brightest in the similarity matrix, reflecting strong genuine matching scores. In contrast, finger 110 has the worst mean NFIQ score (3.75), indicating lower-quality fingerprints. This is consistent with its lowest genuine matching scores and the darkest block in the bottom-right corner of the matrix.

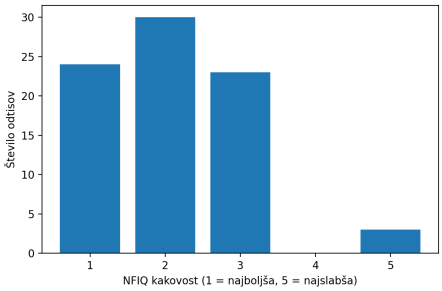



Fig. 3. NFIQ quality of the 80 images (1 best, 5 worst).

The EER threshold is $T = 12$ (FAR 10.7%, FRR 10.4%) (Fig. 4). The accuracy-optimal threshold is $T = 26$, with accuracy 97.94% (about 65 wrong pairs), compared with 91.1% for predicting "impostor" for all pairs.

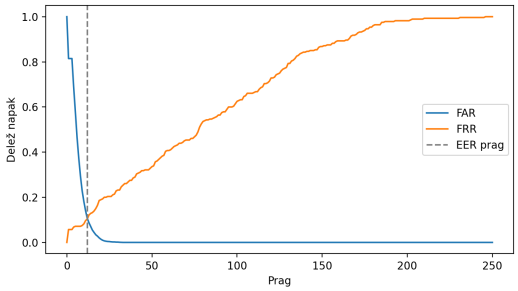



Fig. 4. FAR and FRR as functions of the Bozorth3 threshold T.

`pcasys` assigned 44 images to type L, 28 to R and 8 to W. Fingers 102 and 103 received the same type for all their impressions, whereas the other eight fingers received different types for different impressions. The filter therefore skipped 1808 pairs, including 88 of the 280 genuine pairs (31%), and rejected all of them. Accuracy fell to 95.92% at $T = 26$ and reached at most 95.98% at $T = 27$, because the predicted type was unstable between impressions of one finger.

## IV. Conclusion

Bozorth3 with a threshold of 26 separates same and different fingers with 97.94% accuracy, and lower image quality coincides with lower genuine scores. Pre-classification with pcasys lowered the accuracy to 95.98%, because inconsistent type predictions removed 31% of the genuine pairs. The results are optimistic, because we chose the threshold and evaluated the pipeline on the same small dataset.